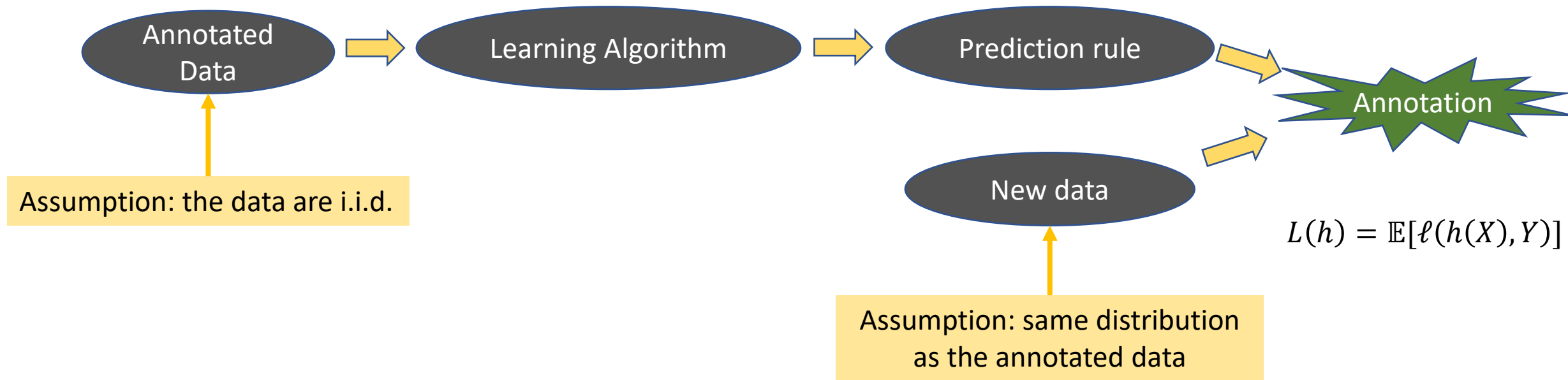


Machine Learning B Summary

Yevgeny Seldin

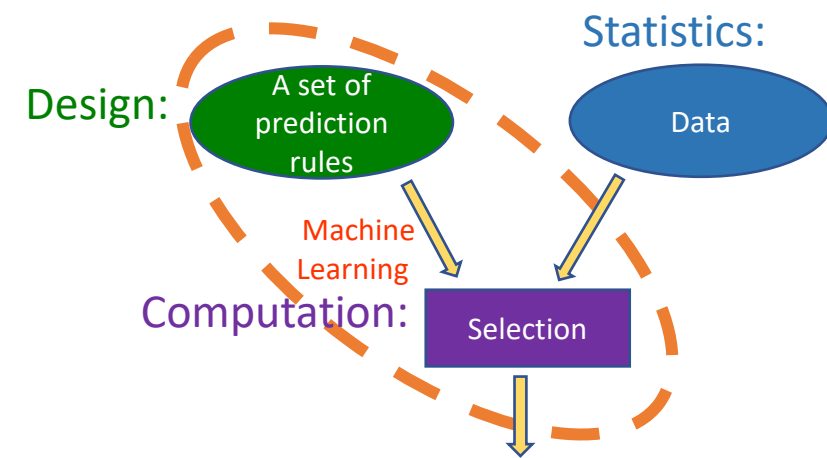
“Classical” Supervised Learning



- We are in the same learning model as in ML-A
- All the same limitations apply (check the “Summary” and “Limitations” lectures from ML-A)
- We only derived tighter bounds + analyzed uncountably infinite selection

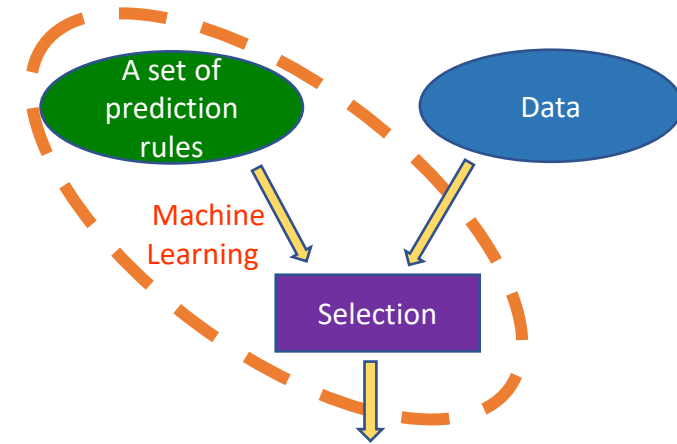
A high-level view

- Learning is a selection process
- Design:
 - Construct a set of options (predictors) to select from
 - Linear separators, decision trees, neural networks, ...
 - Prior knowledge and domain knowledge are helpful
- Statistics:
 - Estimate the quality of various predictors
 - Estimates based on the data are random samples of the true quality
 - The true quality is typically unknown
 - Exploit prior knowledge for tighter estimation
- Computation:
 - Efficiently select “good” predictors out of a large (potentially infinite) set
- All the 3 elements should work together!



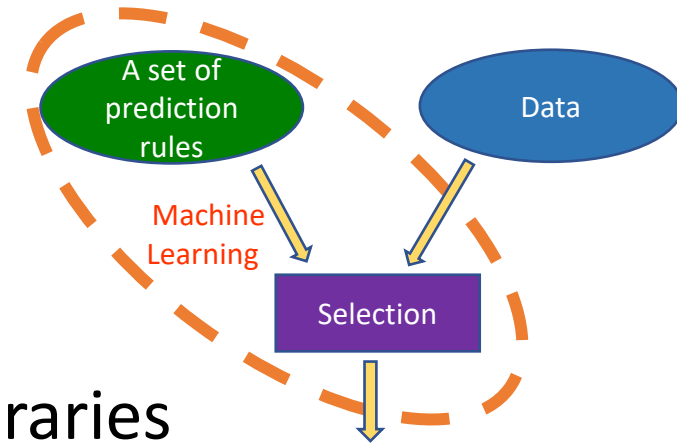
Majority Votes

- Combine predictions of multiple classifiers
- Involves selection of classifiers (potentially “soft” selection and weight tuning)
 - And so, design, statistics, and computation
- Estimate the ensemble quality
 - Statistics



ML Libraries vs. Users

- Significant selection may be happening outside ML libraries
 - E.g., selection of libraries and their initialization
- ML libraries can sometimes (not always!) handle the internal selection
- External selection must be handled by the user



The Statistical Side

Summary of the bounds

Bounds on Moment Generating Functions (MGF)

- kl-Lemma: $\mathbb{E}[e^{nkl(\hat{p}_n||p)}] \leq 2\sqrt{n}$
- Hoeffding-Lemma: $\mathbb{E}[e^{\lambda(X - \mathbb{E}[X])}] \leq e^{\frac{\lambda^2}{8}}$
- Bernstein's Lemma: ...
- Unexpected Bernstein's Lemma:

Change of Measure (CoM)

$$\mathbb{E}_\rho[f(h)] \leq KL(\rho||\pi) + \ln \mathbb{E}_\pi[e^{f(h)}]$$

Chernoff+Markov

Bounds for a single prediction rule (no selection)

$$\mathbb{P}(d(\hat{L}(h, S), L(h)) \geq \varepsilon) = \mathbb{P}(e^{d(\hat{L}(h, S), L(h))} \geq e^\varepsilon) \leq e^{-\varepsilon} \mathbb{E}[e^{d(\hat{L}(h, S), L(h))}]$$

PAC-Bayesian bounds

$$\begin{aligned} \mathbb{P} \left(\exists \rho: \mathbb{E}_\rho[d(\hat{L}(h, S), L(h))] \geq KL(\rho||\pi) + \ln \frac{\text{bound} \left(MGF \left(d(\hat{L}(h, S), L(h)) \right) \right)}{\delta} \right) \\ \stackrel{\substack{\text{MGF} \\ \text{bound}}}{\leq} \mathbb{P} \left(\exists \rho: \mathbb{E}_\rho[d(\hat{L}(h, S), L(h))] \geq KL(\rho||\pi) + \ln \frac{\mathbb{E}_\pi \left[\mathbb{E}_S \left[e^{d(\hat{L}(h, S), L(h))} \right] \right]}{\delta} \right) \\ \stackrel{\substack{\text{CoM}}}{\leq} \mathbb{P} \left(\mathbb{E}_\pi \left[e^{d(\hat{L}(h, S), L(h))} \right] \geq \frac{\mathbb{E}_S \left[\mathbb{E}_\pi \left[e^{d(\hat{L}(h, S), L(h))} \right] \right]}{\delta} \right) \\ \stackrel{\substack{\text{Markov}}}{\leq} \delta \end{aligned}$$

Oracle bounds for the weighted majority vote (WMV)

$$\begin{aligned} L(MV_\rho) &\leq 2\mathbb{E}_{h \sim \rho}[L(h)] \\ L(MV_\rho) &\leq 4\mathbb{E}_{(h, h') \sim \rho^2}[L(h, h')] \end{aligned}$$

Special case if ρ is a delta-distribution

Union bound

$$\mathbb{P}(\exists h \in \mathcal{H}: \dots) \leq \sum_{h \in \mathcal{H}} \mathbb{P}(\dots)$$

Projection + union bound

VC Analysis

Occam's razor

VC Analysis of SVMs

PAC-Bayes Analysis of SVMs

PAC-Bayesian bounds for the WMV

PAC-Bayes vs. VC vs. VC+Occam

- PAC-Bayes (Pinsker's relaxation)

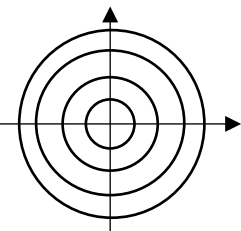
$$\mathbb{E}_\rho[L(h)] \leq \mathbb{E}_\rho[\hat{L}(h, S)] + \sqrt{\frac{\text{KL}(\rho||\pi) + \ln \frac{2\sqrt{n}}{\delta}}{2n}}$$

- VC (Empirical Risk Minimization – ERM)

$$L(h) \leq \hat{L}(h, S) + \sqrt{\frac{8 \ln \frac{2((2n)^{d_{VC}(\mathcal{H})} + 1)}{\delta}}{n}}$$

- VC+Occam (Structural Risk Minimization – SRM)

$$L(h) \leq \hat{L}(h, S) + \sqrt{\frac{8 \ln \frac{2((2n)^{d_{VC}(\mathcal{H}_{j(h)})} + 1)}{\pi(j)\delta}}{n}}$$



Special case: SVMs

$$L_{\text{FAT}}(h_{w,b}) \leq \hat{L}_{\text{FAT}}(h_{w,b}, S) + \sqrt{\frac{8 \ln \frac{2((2n)^{\lceil \|w\|^2 \rceil + 1} + 1) \lceil \|w\|^2 \rceil (\lceil \|w\|^2 \rceil + 1)}{\delta}}{n}}$$

- PAC-Bayes

- Complexity $\pi(h)$ defined for each h individually
- Penalization by the actual selection $\text{KL}(\rho||\pi)$
 - No selection, no penalty
 - $\rho = \pi \Rightarrow \text{KL}(\rho||\pi) = 0$
- Possibility to incorporate prior knowledge via $\pi(h)$

- VC

- Complexity $d_{VC}(\mathcal{H})$ defined for all $\mathcal{H}$
- “Pre-paid” penalization $d_{VC}(\mathcal{H})$
 - Same cost irrespective of selection
- No possibility to incorporate prior knowledge

- VC+Occam: in-between

- Occam provides partial adaptivity across subsets of $\mathcal{H}$
- $d_{VC}(\mathcal{H}_j)$ dominates the complexity
- $\pi(j)$ allows to incorporate structural prior knowledge, but only limited data-dependent prior knowledge

The Computational Side

Two ways of convexifying learning problems

- Convexify the loss function (surrogate loss)
 - Typically used for ERM and SRM
 - E.g. hinge loss in SVMs
 - Minimizes the surrogate loss
 - The gap to the minimizer of the original loss may be large
- Work in the space of distributions on $\mathcal{H}$
 - PAC-Bayes: $\mathbb{E}_{\rho}[L(h)] \leq \frac{\mathbb{E}_{\rho}[\hat{L}(h, S)]}{1 - \frac{\lambda}{2}} + \frac{\text{KL}(\rho || \pi) + \ln \frac{n+1}{\delta}}{n\lambda(1 - \frac{\lambda}{2})}$
 - Not necessarily jointly convex, but amenable to alternating minimization
 - Works with the original loss function, but provides a randomized classifier
 - Can be derandomized via majority vote
- Sometimes we use both

